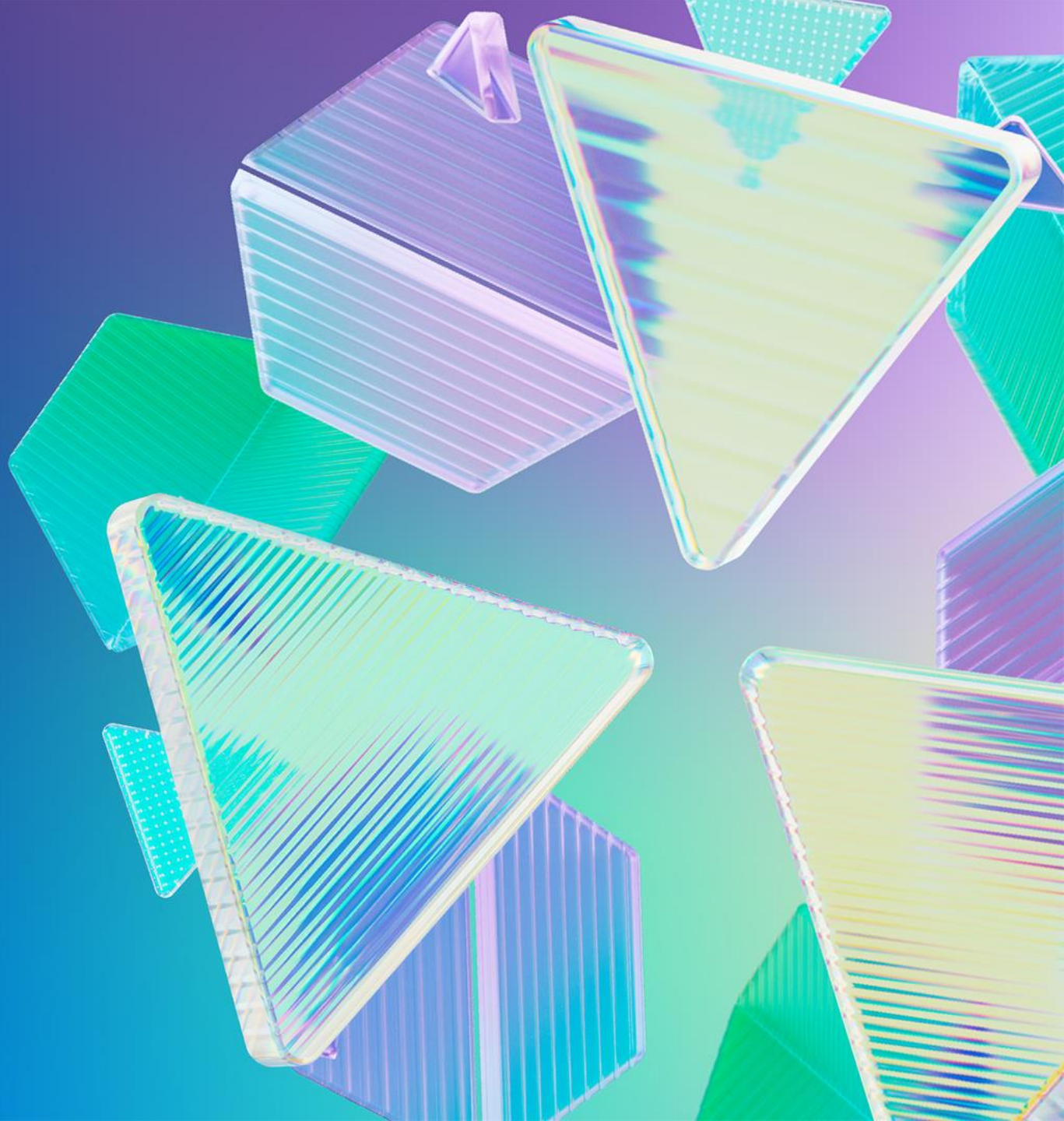
Microsoft

MCAPS Tech Connect



Microsoft

Agentic Platform Engineering with GitHub Copilot

Ray Kao – Developer Global Black Belts
Diego Casati – Migrate & Modernize Global Black Belts

GitHub Copilot SDK Enterprise Challenge

Feb 9 – Mar 7

Showcase customer-oriented, enterprise grade MVPs built with the GitHub Copilot SDK

Why now?

- Copilot started as an "in-the-IDE" assistant that helps to write code faster **-> the SDK lets customers embed the same agentic core that powers Copilot CLI into their own apps.**
- We're kicking off a three-week challenge to build and showcase the most compelling, production-ready solutions for the enterprise

Audience & Timeline

The challenge is open to **all MCAPS FTEs.** You may join as individuals or team up to 3 people.

Important Dates

- **Kickoff:** Feb 9th
- **Office Hours:** Week of Feb 9th
- **Submission Deadline:** Mar 7th at 10 PM PST
- Winners will be announced in Apr

Check out our landing page

Aka.ms/FY26SDKChallenge

Submit your entry through the survey form by Mar 7th 10PM PST

Submission Form

What you'll submit

- A **short summary** of your project
- A **video** of the solution in action (3 mins max)
- Working solution with **code in a GitHub repo**
- **Presentation deck** with 1-2 slides

For additional **bonus points**, you may also include:

- Product Feedback on GHCP SDK
- Validation with customer

The Loot

- Top 3 Finalists will present to Engineering Leaders
- GitHub Swag: 1st place - $250, 2nd place - $150, 3rd place - $100
- Credly Badge
- Internal and External Amplification (Viva/LinkedIn)
- Coaching session with Engineering / Business leaders

Classified as Microsoft Confidential

Agenda

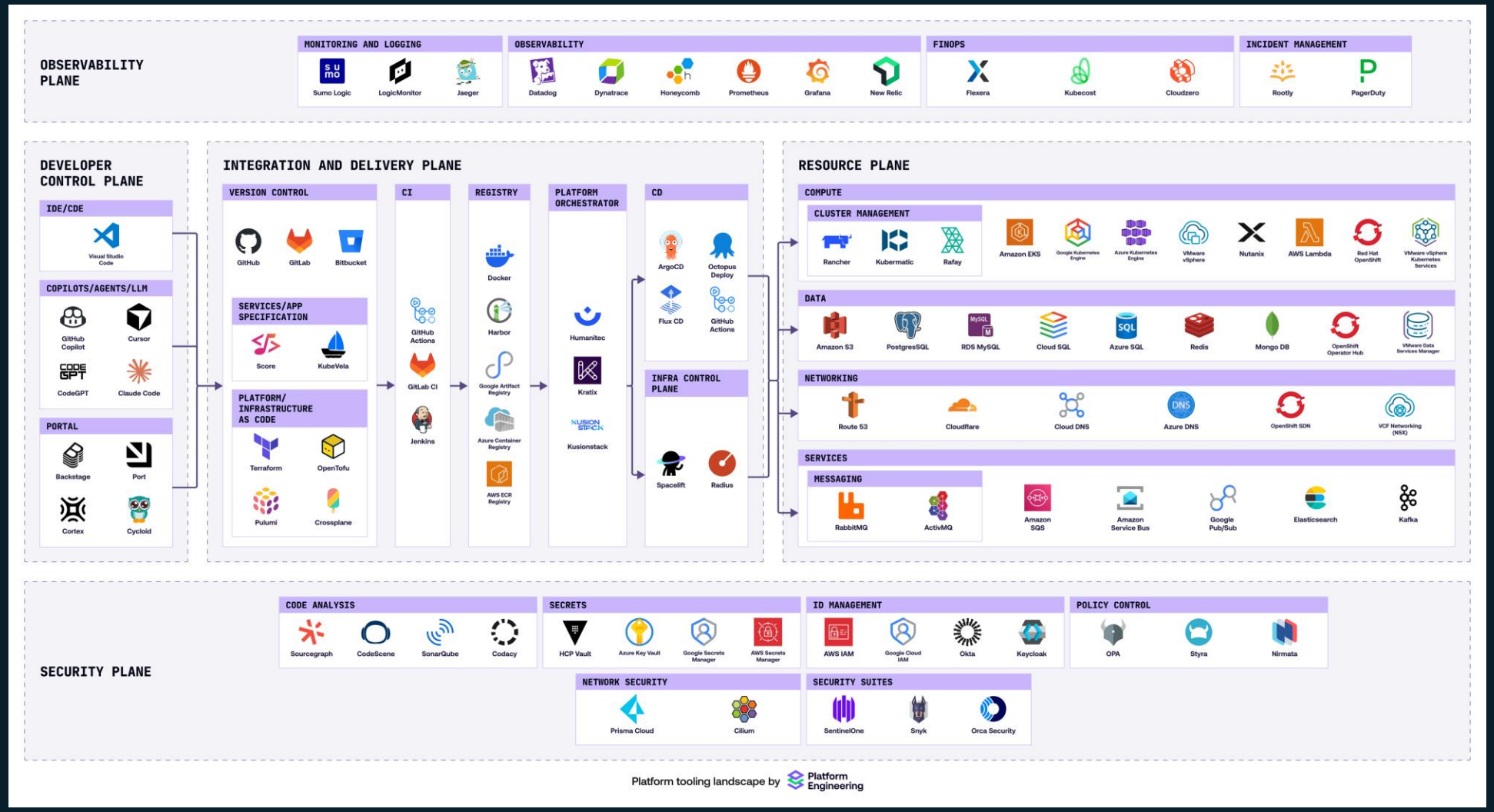
- Goals
- Act One: Platform-aware assistance
- Act Two: Agentic enforcement of standards
- Act Three: Agent-assisted operations
- Final takeaways & next steps

https://aka.ms/agentic-platform-engineering

Session Abstract

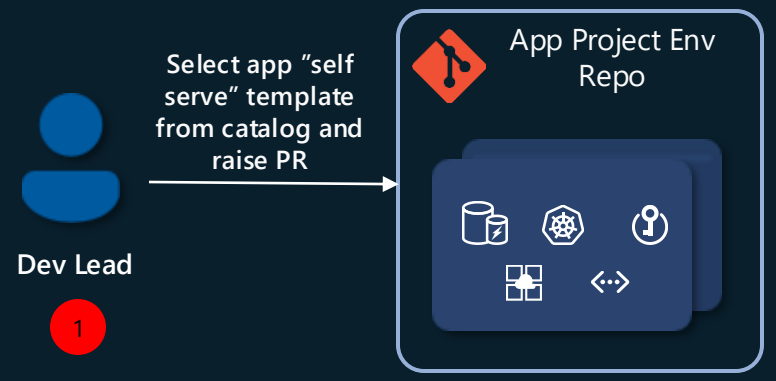
Platform engineering teams must deliver secure, scalable, self-service platforms while supporting more teams and environments. This session shows how GitHub Copilot, Microsoft Foundry, and AKS enable agentic patterns that reduce cognitive load, enforce standards, and help platforms evolve sustainably.

Platform Engineering Landscape



Platform Tooling | Platform Engineering

Case in point: Platform Engineering on AKS with GitOps, CAPZ, and ASO



Build a GitOps-Driven Platform on AKS with the App of Apps Pattern | AKS LABS

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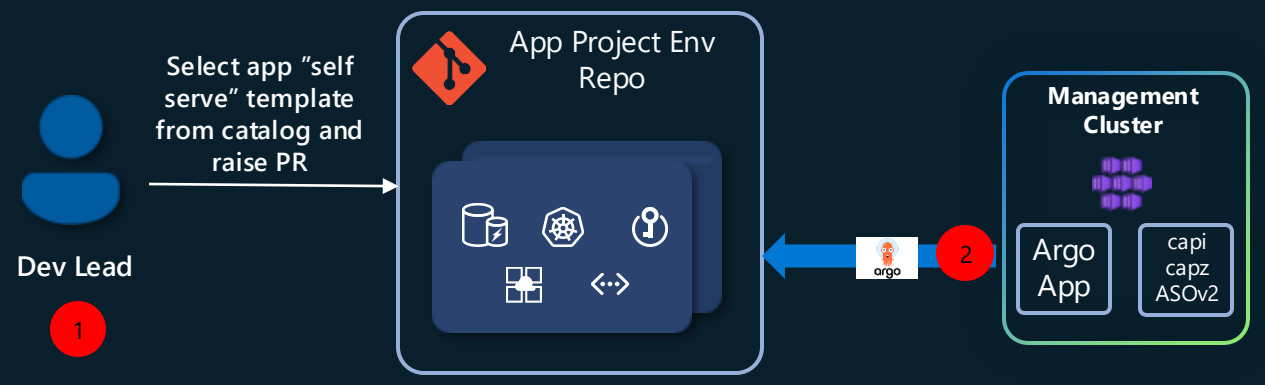

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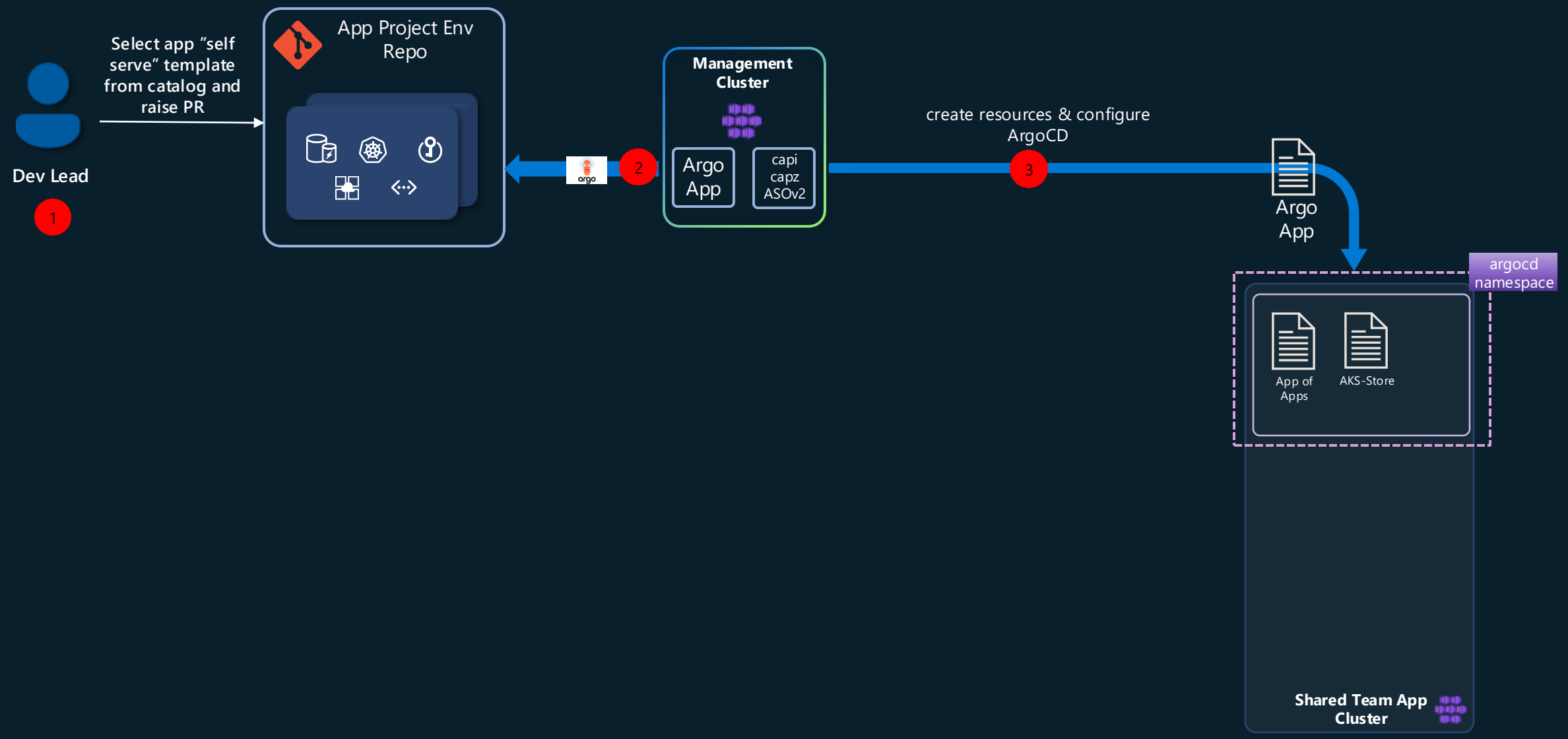

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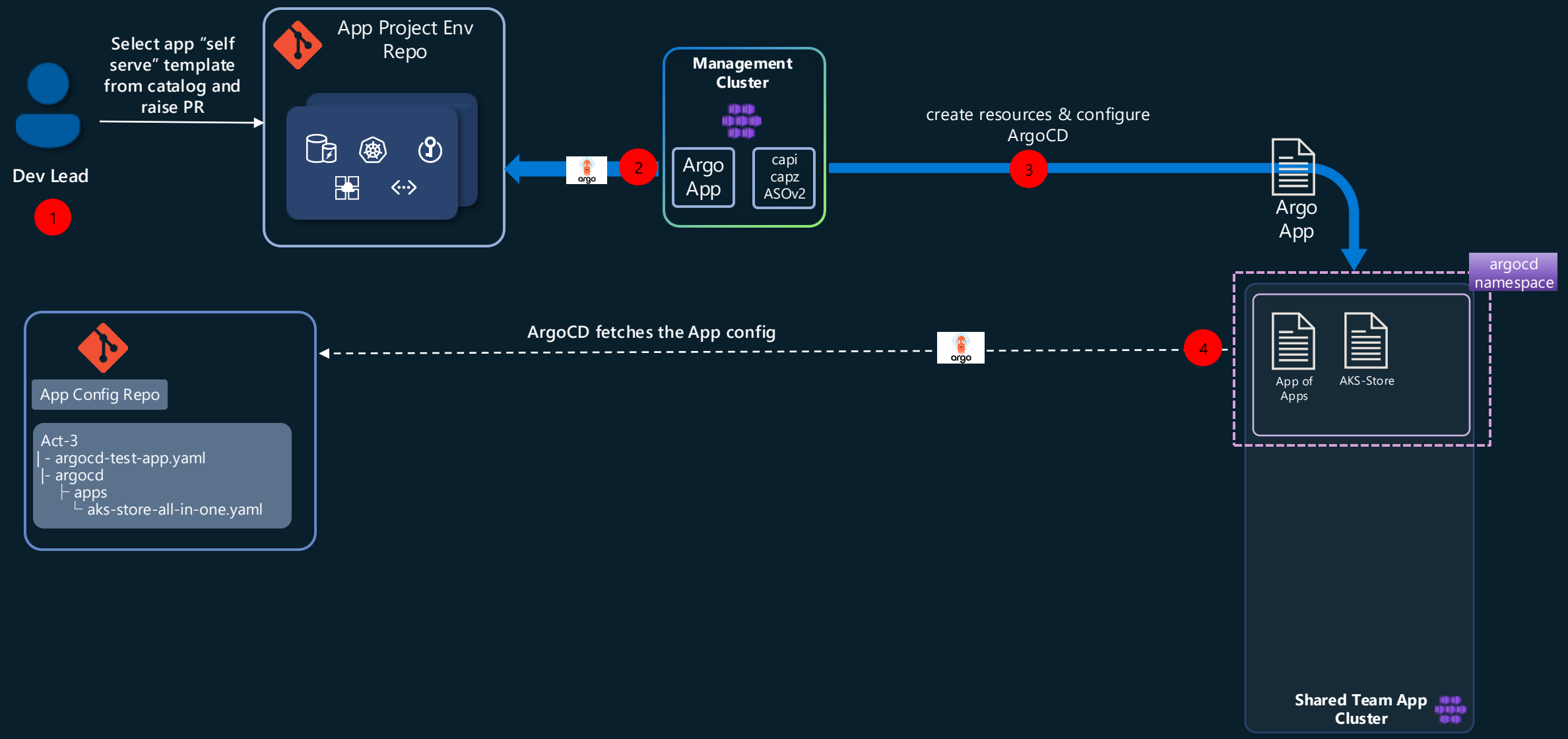

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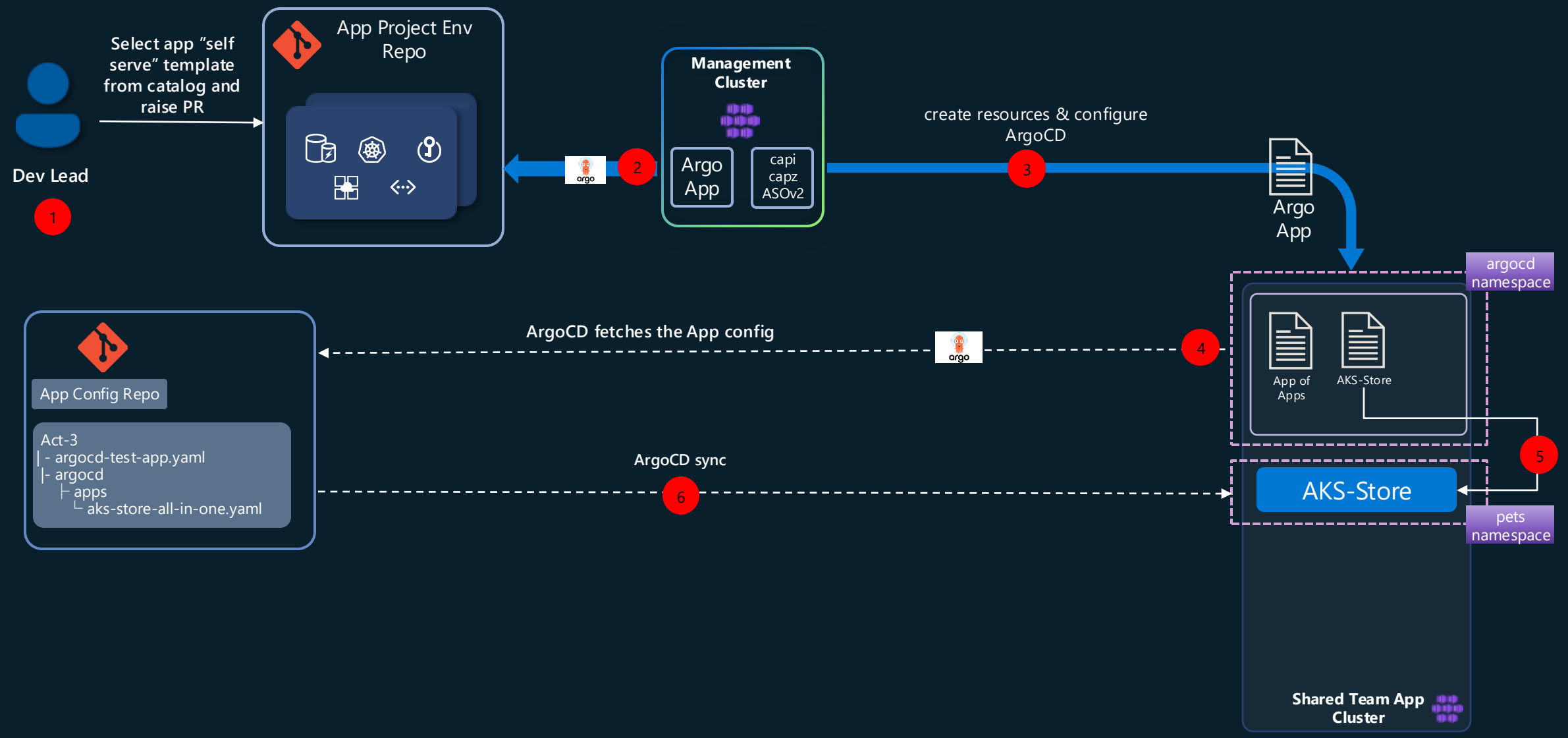

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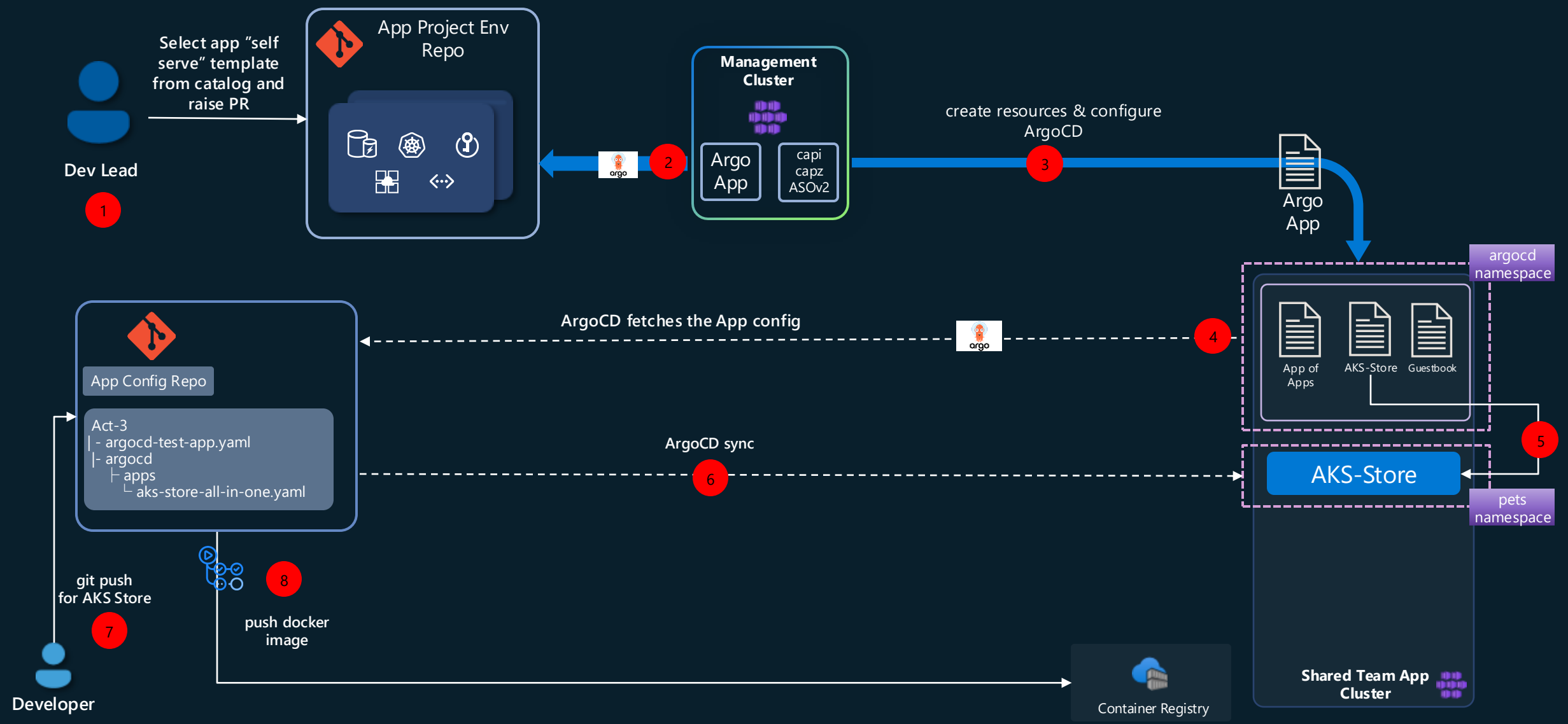

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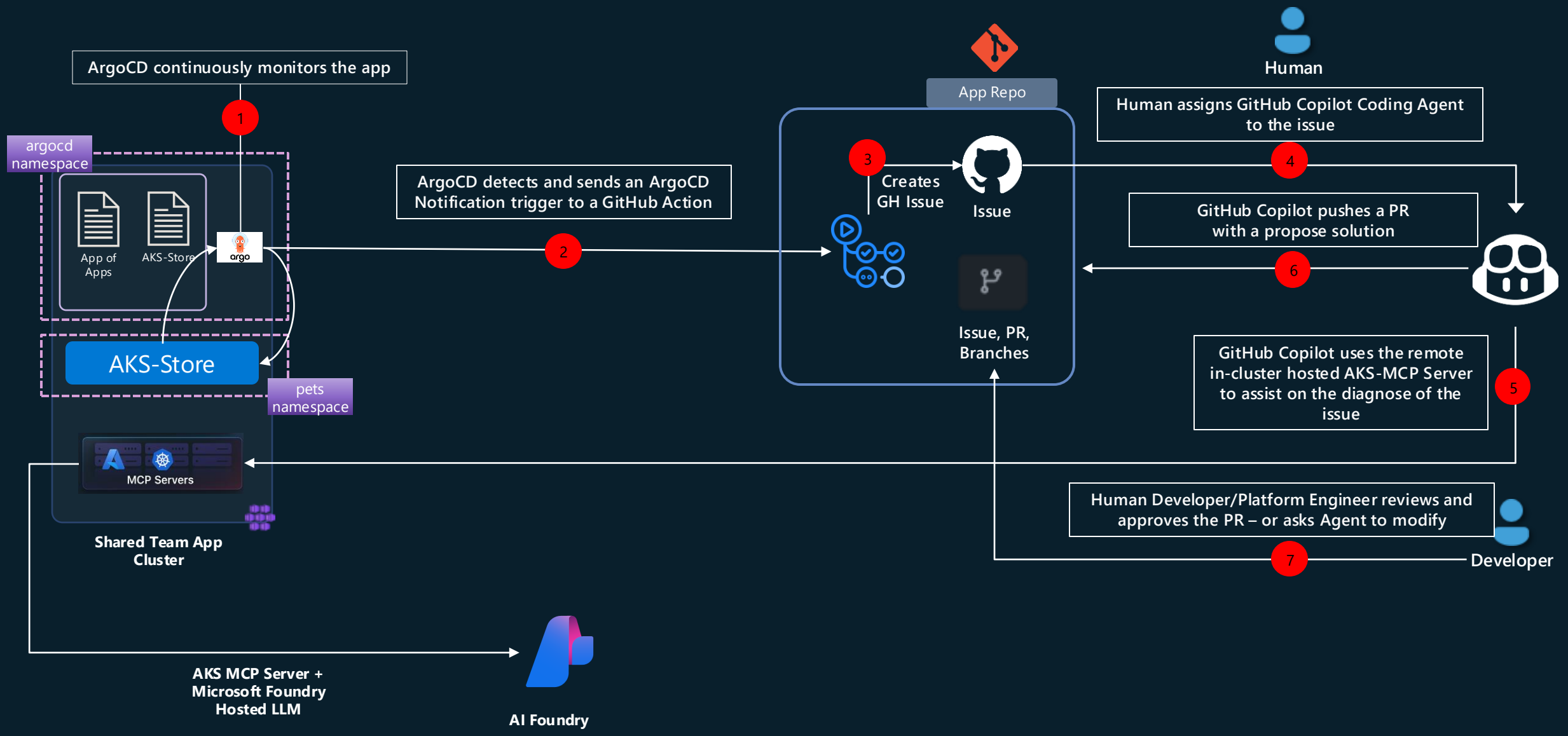
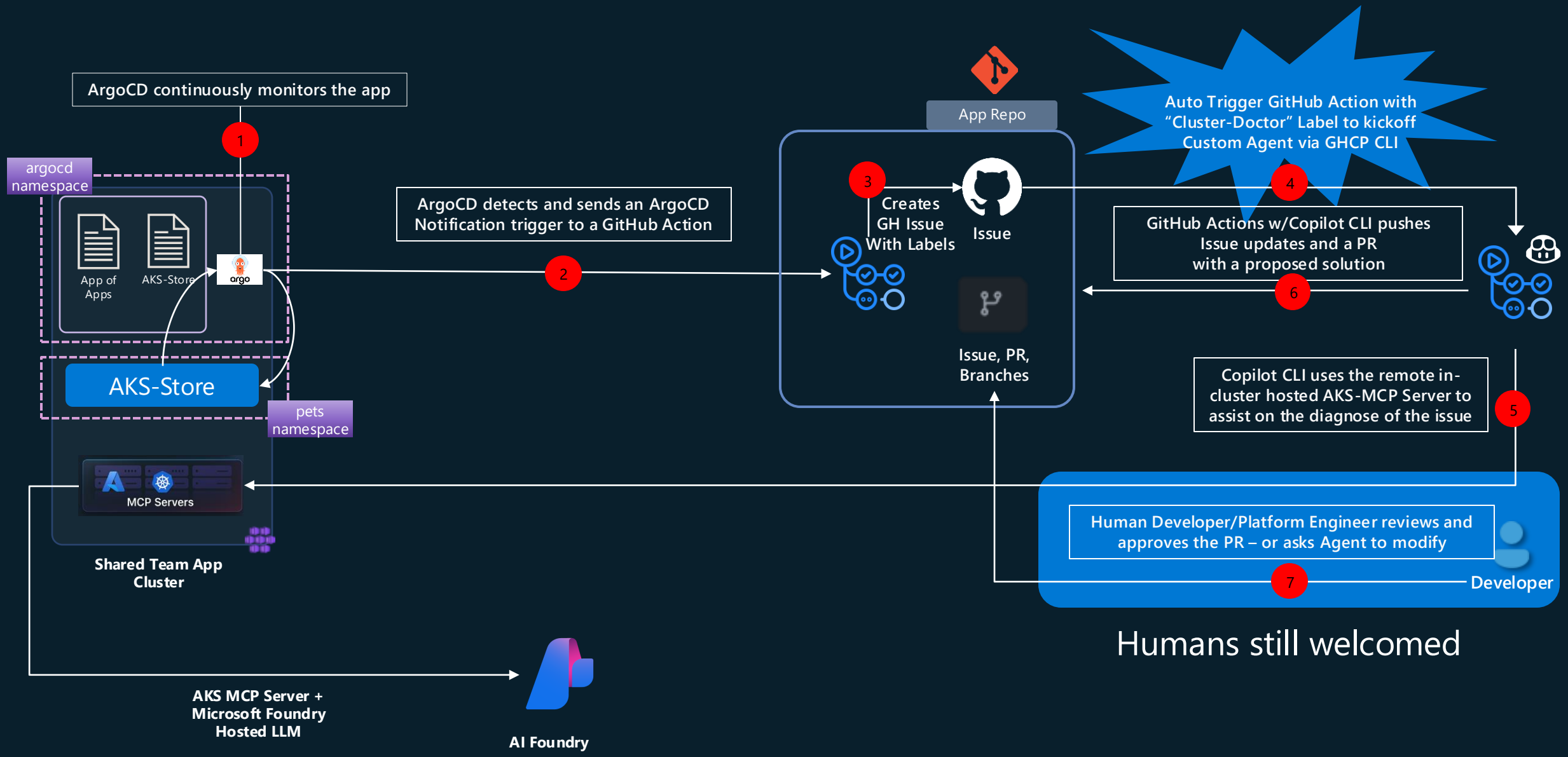
Agentic Platform Engineering



Agentic Platform Engineering – draft.v2.final.test.sp2



Case in point: Platform Engineering on AKS with GitOps, CAPZ, and ASO

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Introduction to AKS Labs
Getting Started
Networking
Security
Storage
Operations
AI
Platform Engineering
Platform Engineering on AKS with GitOps, CAPZ, and ASO
Build a GitOps-Driven Platform on AKS with the App of Apps...
Expanding the platform capabilities with Kube Resource...
Migration

Platform Engineering › Platform Engineering on AKS with GitOps, CAPZ, and ASO

Platform Engineering on AKS with GitOps, CAPZ, and ASO

Welcome to the **Platform Engineering on AKS** workshop. This hands-on workshop will guide you through implementing a robust platform engineering environment using the Cloud Native pattern on Azure Kubernetes Service (AKS). The environment leverages GitOps practices and integrates tools such as ArgoCD, and the Cluster API Provider for Azure (CAPZ) along with Azure Service Operator (ASO). By the end of this workshop, participants will be able to deploy infrastructure and application environments using these tools.

Objectives

- Provide a foundational understanding of key tools in the platform engineering space
- Build a control plane AKS cluster using Azure CLI
- Bootstrap GitOps with ArgoCD
- Demonstrate how CAPZ and ASO can provision infrastructure and Kubernetes clusters
- Show how to deploy new resources in Azure, using CAPI/CAPZ to generate a new AKS cluster,

> **TIP**
> If you have used Infrastructure as Code tools like `terraform` to create resources in Azure, you can use the `asoctl` tool to convert these Azure resources into ASO deployments. To find out more about `asoctl` take a look here. You can download `asoctl` here

---

Prerequisites

- Azure Subscription
- Azure CLI version 2.60.0 or later
- kubectl version 1.28.9 or later

Objectives
Prerequisites
Components
Infrastructure as Code
Architecture Overview
Step 1: Create the AKS cluster
Step 2: Create a user-assigned managed identity for CAPZ
Step 3: Install ArgoCD
Step 4: Access ArgoCD UI
Step 5: Install Cluster API Provider for Azure (CAPZ)
Sample 1: Create a new AKS cluster as an Argo CD Application
Sample 2: Create new ASOv2 resources
Troubleshooting Guide
CAPI Provider Version Mismatches
CRD Conflicts During Reinstallation
Cluster Creation Stuck in Provisioning
CAPZ Provider Not Installing
Summary

Build a GitOps-Driven Platform on AKS with the App of Apps Pattern | AKS LABS

Case in point: Platform Engineering on AKS with GitOps, CAPZ, and ASO

Build a GitOps-Driven Platform on AKS with the App of Apps Pattern | AKS LABS

Platform Engineering Acts

Act One: Platform-aware assistance

Act Two: Agentic enforcement of standards

Act Three: Agent-assisted operations

Act One —

The Platform Is Growing Faster Than the Team

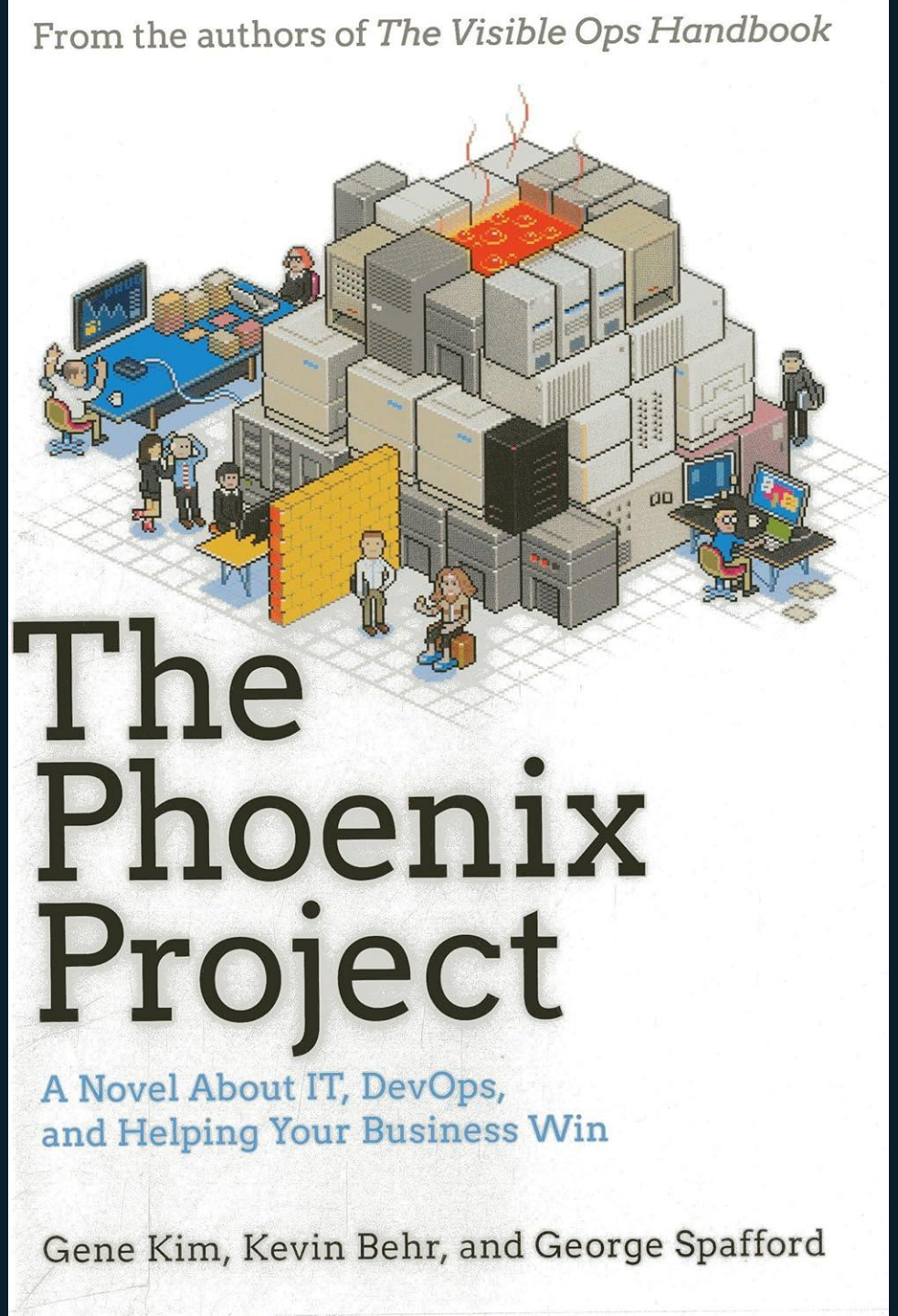
Act One · Scenario / Problem

- App teams onboard faster than platform support
- Each service needs infra, CI/CD, k8s config, security reviews, docs
- Self-service still needs chats/tickets; bottleneck

Repo

Act One · Why This Is a Problem

- Knowledge lives in people, not systems
- Templates and docs drift quickly
- Copy/paste without understanding guardrails
- Platform team answer and support the same questions repeatedly



Pattern:
Platform-aware assistance at the point of work

Move platform knowledge into the developer workflow so engineers get guidance where they build—not after they break things.

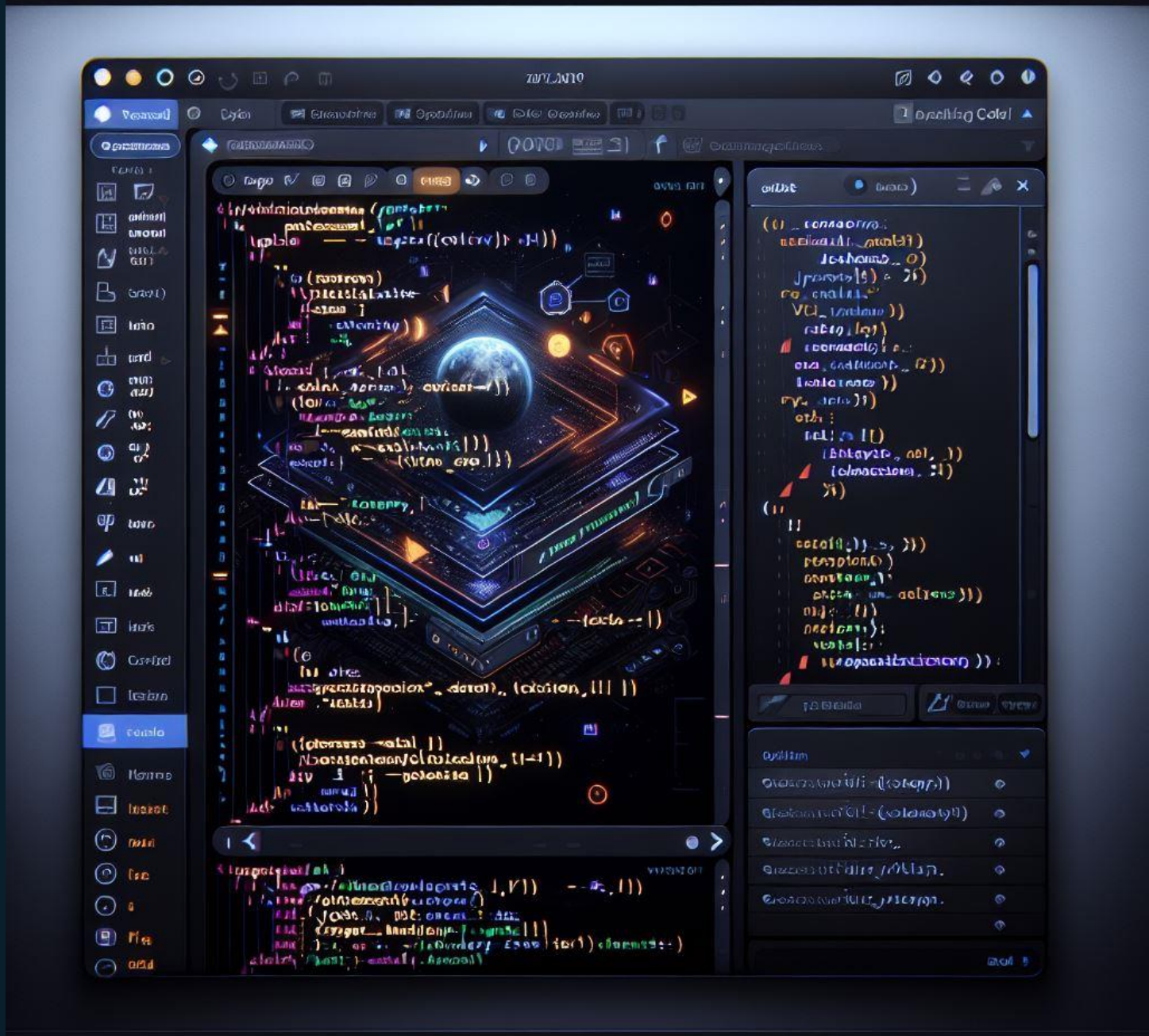
Copilot as a Platform-aware Assistant (VS Code)

- Use Copilot as a platform guide
- Generate Terraform/Helm/Kubernetes manifests to standards
- Explain the ‘why’ in real time
- Shift platform teams from ticket-handlers (execution) to designers (intent).

- Examples (Hat Tip to Ricardo Covo)
  - GitHub: ricardocovo/iac-module-catalog
  - GitHub: ricardocovo/ghcp-infra-reverse-engineer

Act Two —

Standards Exist, but They're Not Enforced



Act Two · Scenario / Problem

- Standards exist but enforcement is inconsistent
- Late reviews find issues after deploy
- Teams unknowingly violate guardrails

Repo

Act Two · Why This Is a Problem

- Humans are poor enforcement mechanisms
- Brittle, hard-coded CI/CD checks
- Rules change faster than pipelines
- Guardrails feel like friction, not enablement

Pattern:
Agentic enforcement with GitHub Copilot & Microsoft Foundry

- Custom Foundry agents that reason, explain and adapt to Enterprise-wide tribal knowledge
- GitHub for PR review & infrastructure human-gated validation
- GitHub Copilot Explain violations & remediation

Foundry IQ

Knowledge bases

Knowledge sources

Agentic RAG

Enterprise access control

Automated data preparation

Sources

Integrations

Knowledge sources

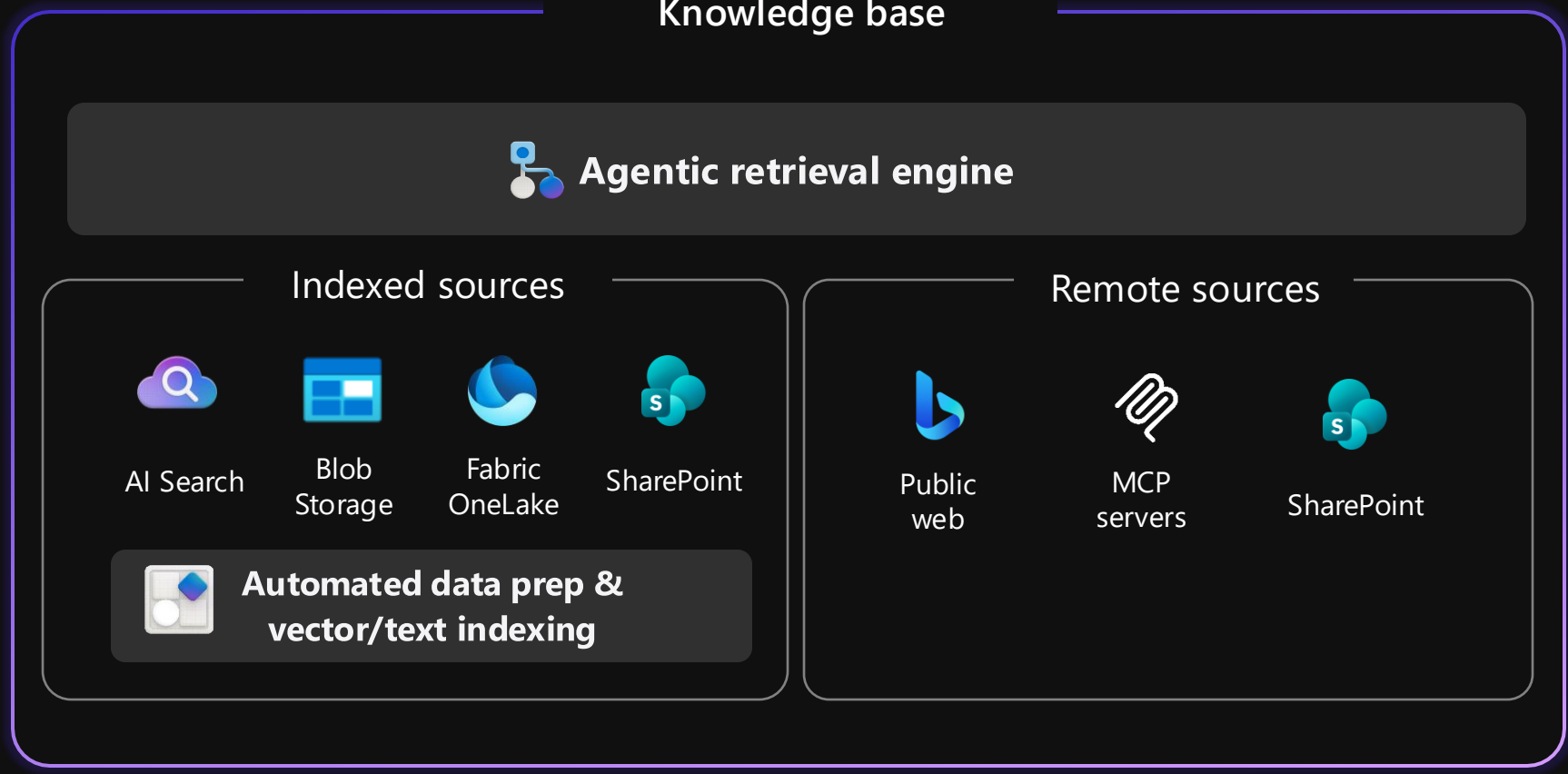
Encapsulate a source of data

- Connection
- Authentication and authorization
- Instructions

Indexed or remote

Indexed sources get automated

- chunking, vectorization,
- OCR/layout-aware parsing with CU
- metadata extraction
- Purview label propagation



This can be a separate knowledgebase workshop on its own…
But not today :’(

Act Three — Kubernetes Operations Don’t Scale Linearly

Act Three · Scenario / Problem

- Kubernetes becomes the operational choke point
- Misconfigurations, failed deployments, runtime issues
- Platform engineers firefight instead of improve platform

Repo

Act Three · Why This Is a Problem

- Kubernetes expertise doesn’t scale across teams
- Runbooks are static (deterministic) and can’t catch everything
- Ops knowledge is reactive, not proactive
- Reliability depends on a few experts

Pattern:
Agent-assisted operations

Use agents to observe, diagnose, and guide — turning operational and tribal knowledge into a shared assistant.

We've got a "Cluster-Doctor" aka. SteveBot

Deconstructing AKS Networking

Steve Griffith – AKS Principal PM
Former Cloud Native GBB

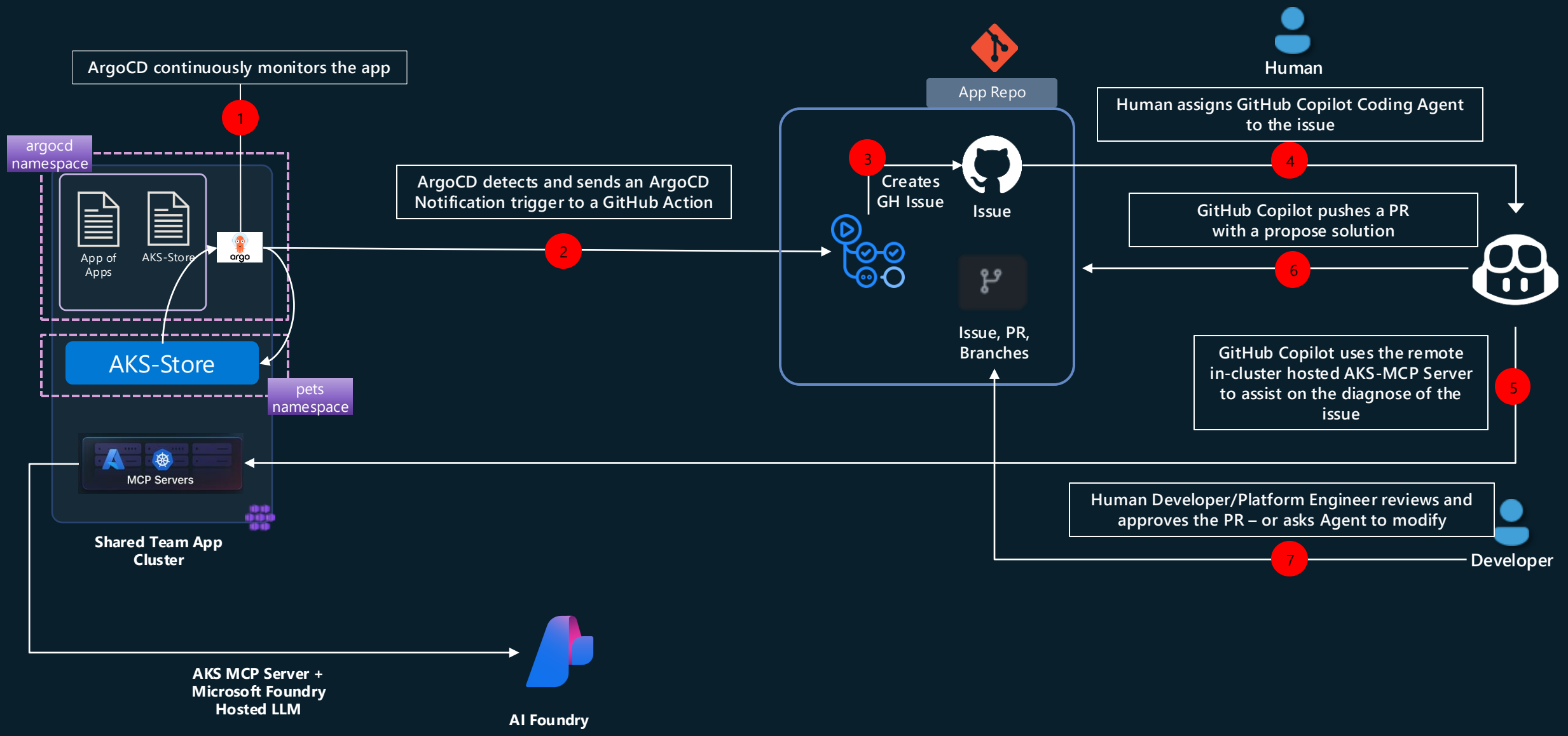
Agentic Platform Engineering

Tools and CLIs

GitHub Copilot

GitHub Actions

Agentic Platform Engineering



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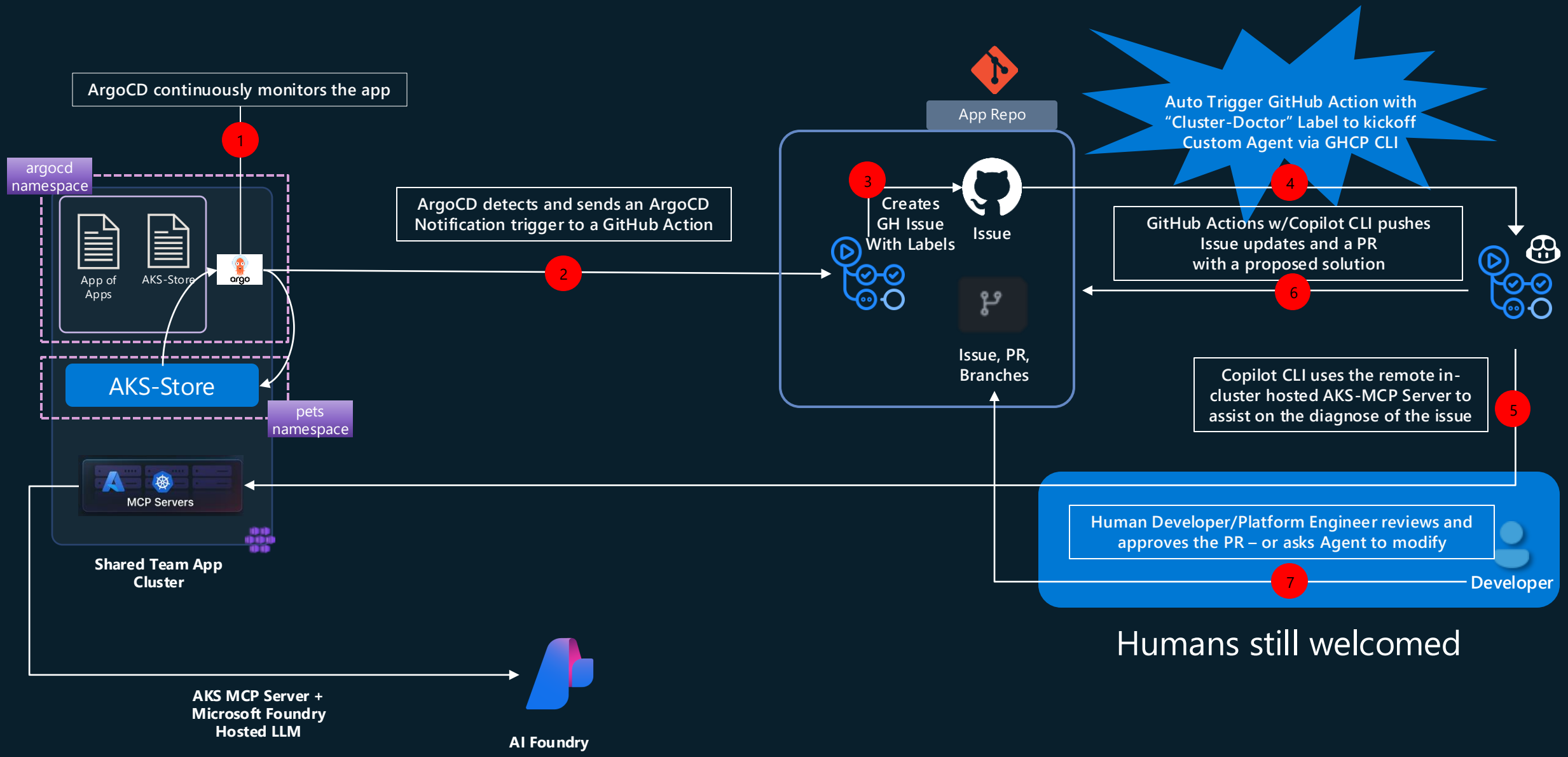


Final Takeaways

Problems repeat even when tools change. So we fix root causes, not just swap tools.

From	**To**
Knowledge bottlenecks	**Platform-aware assistance**
Manual enforcement	**Agentic guardrails**
Ops overload	**Agent-assisted ops**

How to Start & Next Steps

- Begin with one high-friction workflow
- Introduce Copilot first
- Add agents where reasoning/enforcement matter
- Reference architectures & example use cases

Technical Talk | Advanced (300) | Migrate and modernize your estate

Building a mature platform foundation and engineering discipline

In Seattle + Online | Will be recorded

Tue, Feb 10

7:15 PM - 8:15 PM EST

Level 3, Room 320

TT130

Learn what it takes to design, build, and run a mature platform engineering practice—starting with consumer needs, adopting a product mindset, enabling automated and autonomous self-service, and embedding governance and compliance. This...

Show more

Tony Box | Microsoft

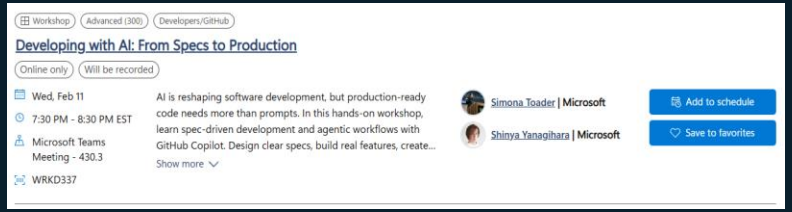
Mike Stiers | Microsoft

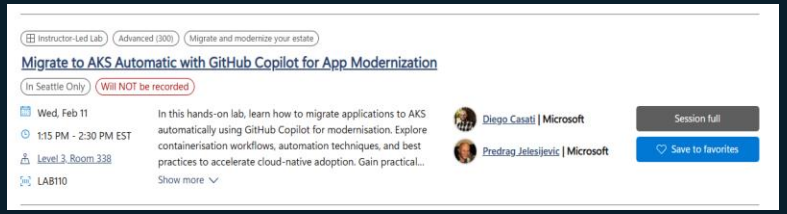
Remove from schedule

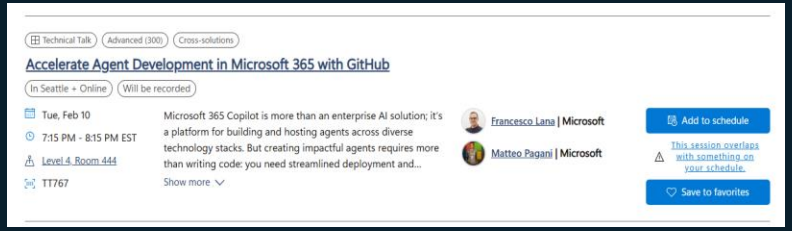
Save to favorites

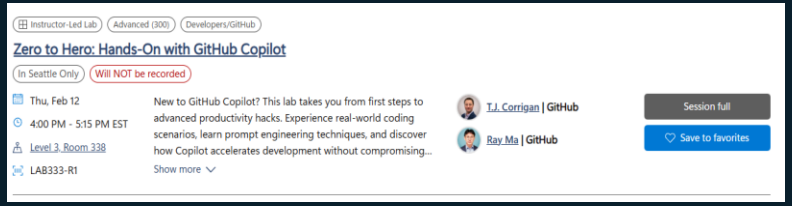
Other Related Sessions

Session Name	Speakers
Hands on with GHCP (Tomorrow)	TJ Corrigan, Ray Ma
Agentic DevOps across GitHub and Azure: Breaking SDLC Silos	Daniel Meppiel
Developing with AI: From specs to Production (as we speak)	Simona Toader, Shinya Yanagihara
Migrate to AKS Automatic with GHCP for App Mod	Diego Casati (who's this Guy???), Predrag Jelesijevic
Vibe Engineering lab: scaling GHCP in the Enterprise	Daniel Meppiel, Chris Romp
Accelerate Agent Development in M365 with GitHub	Francesco Lana, Matteo Pagani









Learning

- Today's Repo:
  - DevExpGbb/agentic-platform-engineering: 🤖 Automated bot/agent for developer workflows

- How to Deploy AKS MCP Server on AKS with Workload Identity | AKS Engineering Blog
- Platform Engineering on AKS with GitOps, CAPZ, and ASO | AKS LABS
- Platform Engineering: Creating Scalable and Resilient Systems | BRK188

https://aka.ms/agentic-platform-engineering

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